

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-304 — Java Applet Programming

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer all sections. Write your answers in your own words as far as practicable.***SECTION — A**

Note: Long-answer questions to be answered in about 750 words each. Attempt any two. [15 × 2 = 30]

Question 1. What are data types and variables in Java? Differentiate between static, instance, and local variables. Why can keywords not be used as variable names? Discuss the **if-else** statement and **switch** statement. Write a program to display the day of the week for a given number using a **switch** statement. (Days: Sunday=0, Monday=1, Tuesday=2, ...)

Question 2. What are arrays? Explain their types and discuss the ways to initialize them.

(a) Write a program to find the maximum element in an array of 20 integer values.

OR

Explain the **for**, **while**, and **do-while** loop structures in Java with examples. What is the difference between entry-controlled loops and exit-controlled loops?

SECTION — B

Note: Short-answer questions to be answered in about 200 words each. Attempt any six. [5 × 6 = 30]

Question 3. How do you read data using the **Scanner** class in Java? State at least three methods defined in the **Scanner** class to read data.

Question 4. What is concatenation of Strings? Write a program to concatenate two strings.

Question 5. Differentiate between the **break** and **continue** statements with examples.

Question 6. What are the features of the Java programming language?

Question 7. Write a program to check whether a number is prime or not.

Question 8. Define JDK and JVM. Why is Java considered a platform-independent language?

Question 9. What is a Constructor? Explain its types.

Question 10. What is an Object and how is it created? Create a class **Student** with at least three attributes and create two objects of that class.

SECTION — C

Note: Answer all objective-type questions.

[10 × 1 = 10]

Question 11. Answer the following:

- (i) Who invented the Java programming language?
 - (a) Guido van Rossum
 - (b) James Gosling
 - (c) Dennis Ritchie
 - (d) Bjarne Stroustrup
- (ii) _____ is used to find and fix bugs in Java programs.
 - (a) JVM
 - (b) JRE
 - (c) JDK
 - (d) JDB
- (iii) What does an Eclipse IDE allow?
 - (a) Editing the program only
 - (b) Compiling the program only
 - (c) Both Editing and Compiling the program
 - (d) Editing, Building, Debugging, and Compiling the program
- (iv) SDK stands for _____.
 - (a) Software Design Kit
 - (b) Serial Development Kit
 - (c) Software Development Kit
 - (d) Serial Design Kit
- (v) Which of the following is not an OOP concept in Java?
 - (a) Polymorphism
 - (b) Inheritance
 - (c) Compilation
 - (d) Encapsulation
- (vi) What will be the output of the following Java program?

```
class Variable_scope {  
    public static void main(String args[]) {  
        int x;  
        x = 5;  
        {  
            int y = 6;  
            System.out.print(x + "_" + y);  
        }  
        System.out.println(x + "_" + y);  
    }  
}
```

- (a) Compilation error
- (b) Runtime error
- (c) 5 6 5 6
- (d) 5 6 5

(vii) What will be the output of the following Java program?

```
class output {  
    public static void main(String args[]) {  
        StringBuffer s1 = new StringBuffer("Quiz");  
        StringBuffer s2 = s1.reverse();  
        System.out.println(s2);  
    }  
}
```

- (a) QuizziuQ
- (b) ziuQQuiz
- (c) Quiz
- (d) ziuQ

(viii) In which memory is a String stored when we create a String using the **new** operator?

- (a) Stack
- (b) String memory
- (c) Heap memory
- (d) Random storage space

(ix) Which keyword is used for accessing the features of a package?

- (a) package
- (b) import
- (c) extends
- (d) export

- (x) Which value is returned by the `read()` method when End of File (EOF) is encountered?
- (a) 0
 - (b) 1
 - (c) -1
 - (d) Null